

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

- A. It activates TLR5 on the surface of macrophages.
- B. It primarily induces the Th1-cell response.
- C. Peptidoglycan is its major antigen that induces an immune response.
- D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think> The patient's symptoms and X-ray findings are consistent with community-acquired pneumonia , which is typically caused by bacterial pathogens such as Streptococcus pneumoniae , Haemophilus influenzae , or Mycoplasma pneumoniae . The patient's response to cephalexin , a broad-spectrum antibiotic , suggests that the causative pathogen is likely a bacterium . Bacteria are typically recognized by the immune system through the activation of Toll-like receptors (TLRs), particularly TLR2 and TLR4 , which recognize peptidoglycan and other bacterial cell wall components . This recognition leads to the activation of innate immune responses , including the production of cytokines and the activation of macrophages . However , the question asks about the immune response elicited by the pathogen , which is more related to the adaptive immune response . The adaptive immune response to bacteria , especially extracellular bacteria like those causing pneumonia , is primarily mediated by Th1 cells . Th1 cells produce cytokines that activate macrophages and promote the killing of intracellular bacteria . The other options are not accurate descriptions of the immune response elicited by the pathogen in this case . TLR5 is activated by flagellin , which is a component of bacterial flagella , but not the primary antigen in this case . Peptidoglycan is indeed a major antigen , but it is not the primary antigen that induces an immune response in this case . Encapsulation is a virulence mechanism used by some pathogens , such as Streptococcus pneumoniae , but it is not the primary mechanism by which this pathogen evades the immune response .</think><answer> B . It primarily induces the Th1-cell response . </answer>